



Manufacturing Engineering I (MEng 3192)

BiT, FMIE, 2016 E.C.
2nd Semester

Manufacturing Engineering I (MEng 3192)

Unit 1

Introduction to Manufacturing

Outlines

- Introduction and overview of manufacturing
- Manufacturing industries and product
- Manufacturing processes
- Selecting manufacturing processes

Introduction and overview of manufacturing

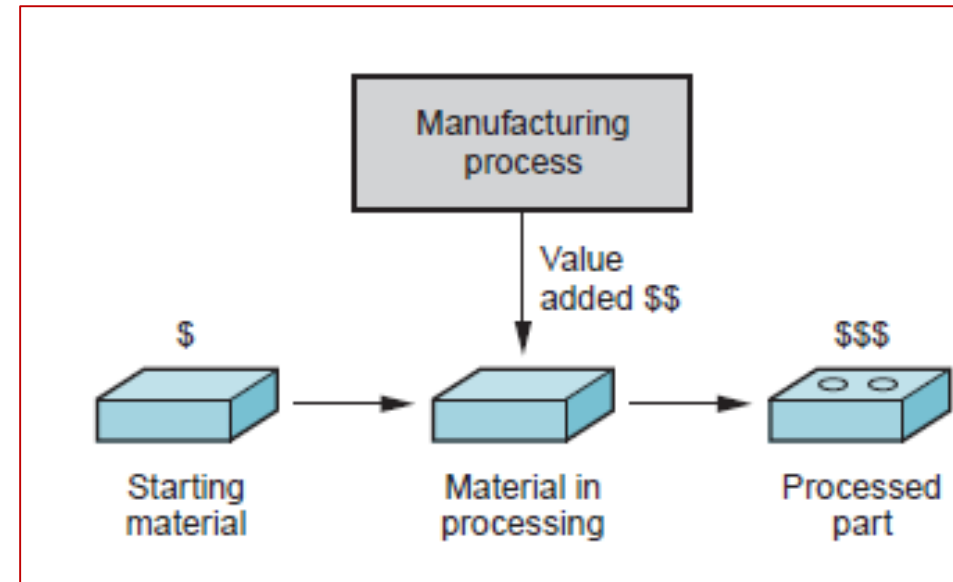
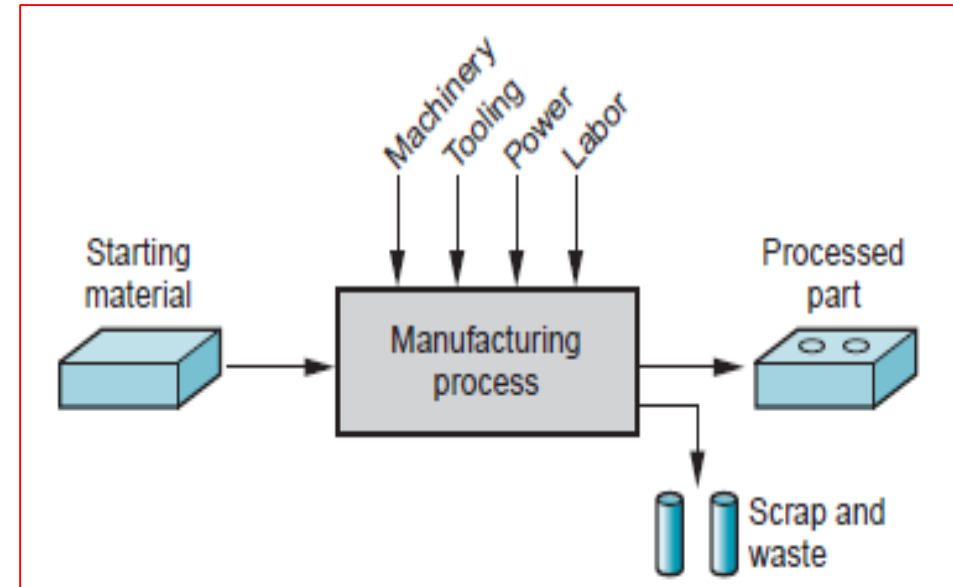
What Is Manufacturing?

- Some objects are made up of one part, such as nails, bolt, nut
- Where as most objects are made from a variety of materials or parts such as **aircraft engines, automobile, ball point pens, bicycles, and computers.**
- All are made by various processes that we call it **manufacturing.**
- Manufacturing is the process of converting **raw materials in to products** and it includes the **design and production** of goods using various **methods and techniques.**



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- Manufacturing is almost always carried out as a **sequence of operations**.
- Each operation brings the material closer to the **desired final state**.



Manufacturing industries and products

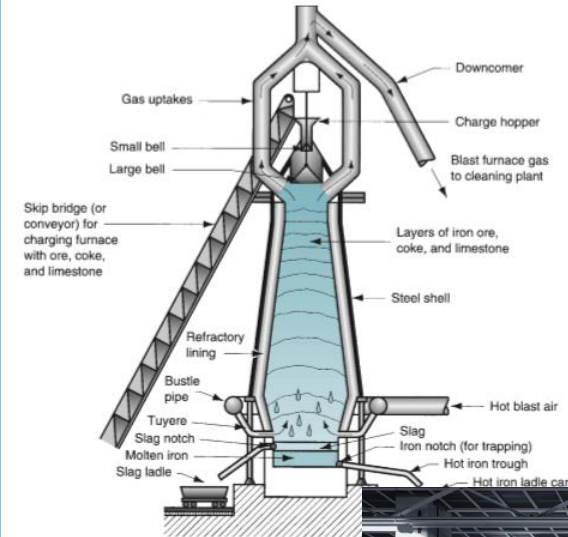
- Manufacturing is an important commercial activity performed by companies that **sell products to customers.**
- The type of manufacturing done by a company depends on the kind of product **it makes.**

Manufacturing Industries

- Industry consists of enterprises and organizations that produce or supply goods and services.
- Industries can be classified as **primary, secondary, or tertiary.**

Manufacturing industries and products

- **Primary industries** cultivate and exploit natural resources, such as **agriculture** and **mining**.
- **Secondary industries** take the outputs of the primary industries and convert them into **consumer and capital goods**.
- Manufacturing is the principal activity in this category, but construction and power utilities are also included.
- **Tertiary industries** constitute the service sector of the economy.



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Specific industries in the primary, secondary, and tertiary categories

Primary	Secondary		Tertiary (Service)	
Agriculture	Aerospace	Food processing	Banking	Insurance
Forestry	Apparel	Glass, ceramics	Communications	Legal
Fishing	Automotive	Heavy machinery	Education	Real estate
Livestock	Basic metals	Paper	Entertainment	Repair and
Quarries	Beverages	Petroleum refining	Financial services	maintenance
Mining	Building materials	Pharmaceuticals	Government	Restaurant
Petroleum	Chemicals	Plastics (shaping)	Health and	Retail trade
	Computers	Power utilities	medical	Tourism
	Construction	Publishing	Hotel	Transportation
	Consumer	Textiles	Information	Wholesale trade
	appliances	Tire and rubber		
	Electronics	Wood and furniture		
	Equipment			
	Fabricated metals			

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Manufactured Products

- Final products made by the manufacturing industries can be divided into two major classes
 - ☞ **Consumer goods and Capital goods,**
- **Consumer goods** are products purchased directly by consumers, such as cars, personal computers, TVs, tires, and tennis rackets.

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Manufactured Products

- **Capital goods** are those purchased by companies to **produce goods** and/or provide services.
- Examples of capital goods include aircraft, computers, communication equipment, medical apparatus, trucks and buses, railroad locomotives, machine tools, and construction equipment.



Manufacturing processes

- A manufacturing process is a designed procedure that results in **physical and/or chemical** changes to a starting work material with the intention of **increasing the value** of that material.
- A manufacturing process is usually carried out as a unit operation, which means that it is a single step in the sequence of steps required to transform the starting material into a final product.

Manufacturing operations can be divided into two basic types:

- 1 Processing operations and
- 2 Assembly operations.

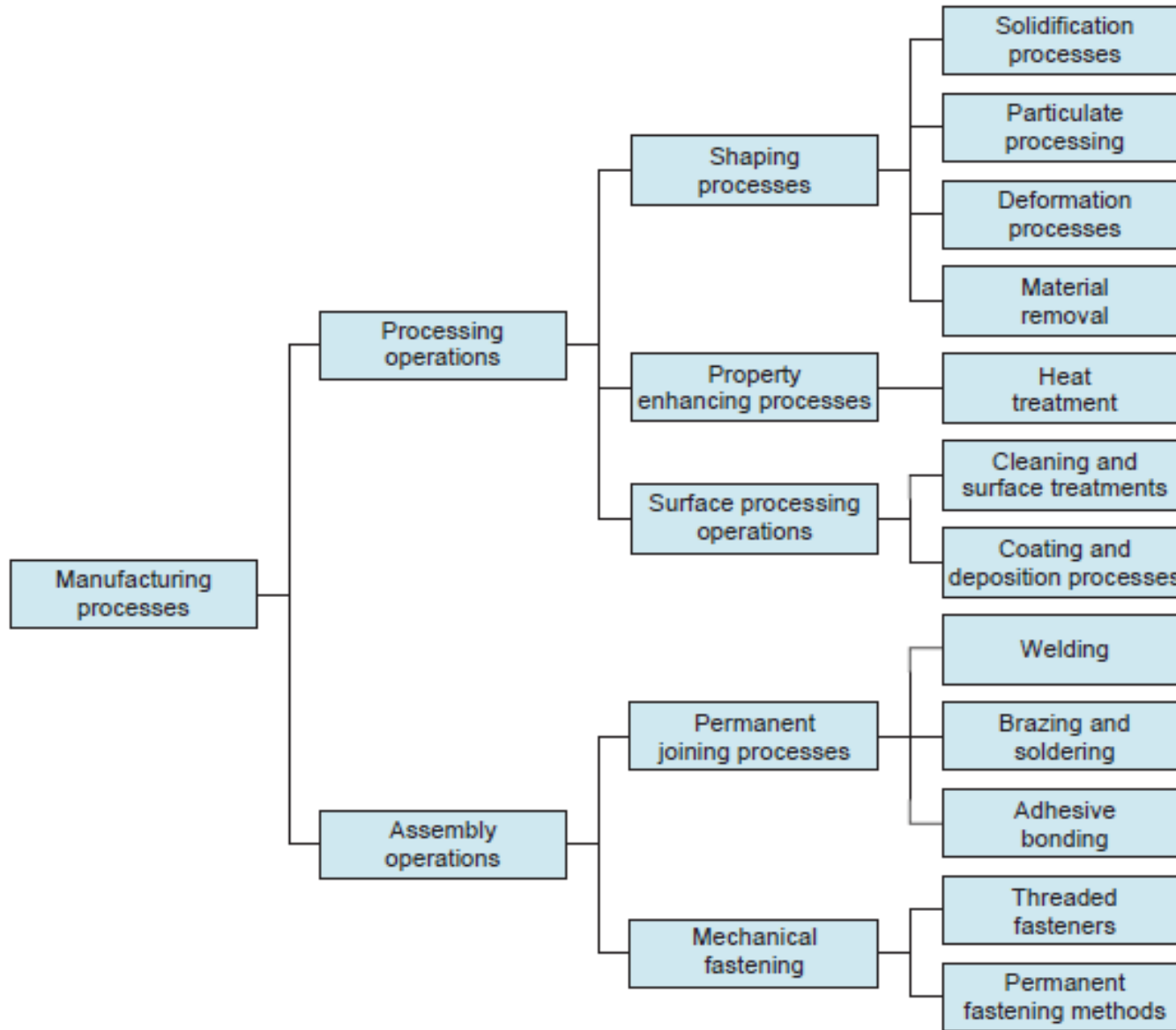
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Processing operation

- Transforms a work material from one state of completion to a more advanced state that is closer to the final desired product.
- It adds value by changing the *geometry, properties, or appearance* of the starting material.
 - **Shaping process**
 - **Property enhancing processes and,**
 - **Surface processing operation**

Assembly operation

- Joins two or more components to create a new entity, called an *assembly, subassembly*, or some other term that refers to the joining process



Selecting manufacturing processes

Broad categories of processing methods for materials are:

- **Casting** processes (expandable and permanent mold)
- **Forming and shaping** processes (rolling, forging, extrusion, drawing , sheet forming)
- **Machining** processes (turning, boring, drilling, milling, planning, broaching ,grinding, ultrasonic machining and chemical, electrical, and electrochemical machining)
- **Joining** processes (welding, brazing, soldering, diffusion bonding, adhesive bonding and mechanical joining)
- **Finishing** processes (honing, lapping, polishing, surface treating, coating)

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- The selection of a particular manufacturing process depends up on not only on the **shape** to be produced but also on a large number of other factors.
- The **type of material and its properties** are basic considerations.
- For example **brittle and hard** materials cannot be formed easily, where as they can be **cast or machined** by several methods therefore the manufacturing process usually alters the properties of materials

